

CSE 332: Computer Architecture and Organization
Assignment#3
Spring 2025

Name: _____

Student ID: _____

Direction: You can consult any resources such as books, online references, and videos for this assignment, however, you have to properly cite and paraphrase your answers when it is necessary. There will be points for partial attempts. You can upload a typed or handwritten copy of your assignment to the canvas. However, you have to show the original copy of your assignment in case of arising questions about the authenticity. You should upload your assignment to canvas by the due date.

Submission guidelines: All our assignments will be based on 100 points so that we can assign weights to get your points for final grade. Whoever fails to submit within the time period assigned through NSU canvas but submit the assignment within the next three days will be punished by 20% deduction of points (e.g. start with -20 points within 100 points). The late submission will be open until semester's end through canvas. After the first 2 days of grace period, the deduction will be 40%.

Note: You are allowed to use only instructions implemented by the actual MIPS hardware provided in the Canvas references. Use assembly language format from the references or the book. Note, base ten numbers are listed as normal (e.g. 23), binary numbers are prefixed with 0b or format such as XX_2 and hexadecimal numbers are prefixed with 0x or format such as XX_{16}/XX_{HEX}

1. **(30 points)** Write a C program and corresponding assembly program based on MIPS ISA that reads three edges for a triangle and computes the perimeter if the input is valid. Otherwise, display that the input is invalid. The input is valid if the sum of every pair of two edges is greater than the remaining edge.
2. **(70 points)** Write an assembly program in MIPS ISA that will take input as your full name- your first and last name, then it will calculate the maximum and minimum of ASCII values (excluding the space between the first and last name) of all the letters of your name and display it in the console.

Run and test your program in CPUlator (MIPS ISA simulator: MIPS32r5 SPIM), submit a printed copy of the program, screenshot the output in the console and result in the integer registers in the Simulator. Also submit your code with .S extension as a separate file.